

Counting Off-Ball Chemistry: Measuring How Often Teammates and Opponents Anticipate Each Other in Elite Soccer

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Introduction. In modern football, elite performance is fundamentally driven by off-ball spatial intelligence. Long before a pass is made, teammates coordinate their runs, create decoy space, and anticipate each other's decisions. Opposing defenders must also anticipate and react to attacking runs before passes arrive. However, standard soccer statistics only count on-ball events like passes, touches, and shots. These box-score numbers miss off-ball teamwork and cannot tell whether two players have genuine chemistry or are simply chasing the ball at the same time. To solve this problem, we borrow a method called Granger causality. Originally developed in economics and widely used in brain network research to track signaling between connected regions, we adapt it to player tracking data. By treating players like connected nodes in a live system, we count how often one player's preceding run directly triggers a teammate's or an opponent's next movement.

Methods. Using 2022 FIFA World Cup optical tracking data collected 5 times per second, we track every player's movement across the entire pitch. We divide each match into rolling 15-second moments of play with a 50% step overlap. Inside every 15-second moment, we test whether Player A's movement triggers Player B's movement one to two seconds later, after accounting for Player B's own momentum. Crucially, the model accounts for the ball in every calculation, which removes the bias of both players simply chasing the ball. Whenever Player A's run directly explains Player B's movement in an active moment, we add one to that pair's anticipation count. Adding these moments across the entire match produces a total Anticipation Count Map for each team (Figure 1). We also apply this method across opposing teams to count how often defenders anticipate attacking runs in the World Cup Final (Figure 2).

Results. Examining tournament contenders across France, Argentina, Morocco, and Serbia reveals clear off-ball tactical patterns:

- **Strikers Triggering Playmakers in Argentina:** In Argentina's 3-0 semi-final win over Croatia, Lionel Messi and Alexis Mac Allister connected across 218 discrete anticipation moments, the highest in the team. Striker Julián Álvarez repeatedly triggered Messi's movement into open attacking space across 169 moments, showing that a hard-working striker's off-ball runs actively trigger the playmaker's positioning.
- **Flank Chemistry in Morocco:** On Morocco's right wing against France, Hakim Ziyech and Achraf Hakimi showed near-identical reciprocal chemistry. Ziyech triggered Hakimi's forward runs 187 times, while Hakimi triggered Ziyech 183 times.
- **Attacking Synergy in Serbia:** In Serbia's 3-3 match against Cameroon, playmaker Dušan Tadić triggering striker Aleksandar Mitrović's box runs was the number one partnership in the entire squad, totaling 215 anticipation moments.
- **Opponent Reading in the Final:** In the World Cup Final between Argentina and France (Figure 2), cross-team counting shows how defenders track attackers. Argentine center-back Cristian Romero anticipated French attackers the most (256 times against Kolo Muani, 196 against Dembélé), while French midfielder Aurélien Tchouaméni read Argentine runners the most (287 times against Messi, 308 against Mac Allister).

Conclusion. Adapting Granger causality gives clubs and coaches a practical way to count off-ball anticipation. Recruitment analysts can check whether a transfer target naturally clicks with current players before buying them, while coaches can identify which opposing defenders regularly anticipate their key attackers to adjust their match plan.

Figure 1: Cumulative Anticipation Event Counts Across Tournament Contenders

Figure 1. Cumulative discrete anticipation event counts for France (WC Final vs Argentina), Argentina (Semi-Final 3-0 vs Croatia), Morocco (Semi-Final vs France), and Serbia (Group Stage 3-3 vs Cameroon). Each cell displays the exact integer count of 15-second tactical phases where the Leader (Row) triggered the Follower (Column) while controlling for ball movement. Outliers and inactive squad members are removed, with zeros explicitly displayed.

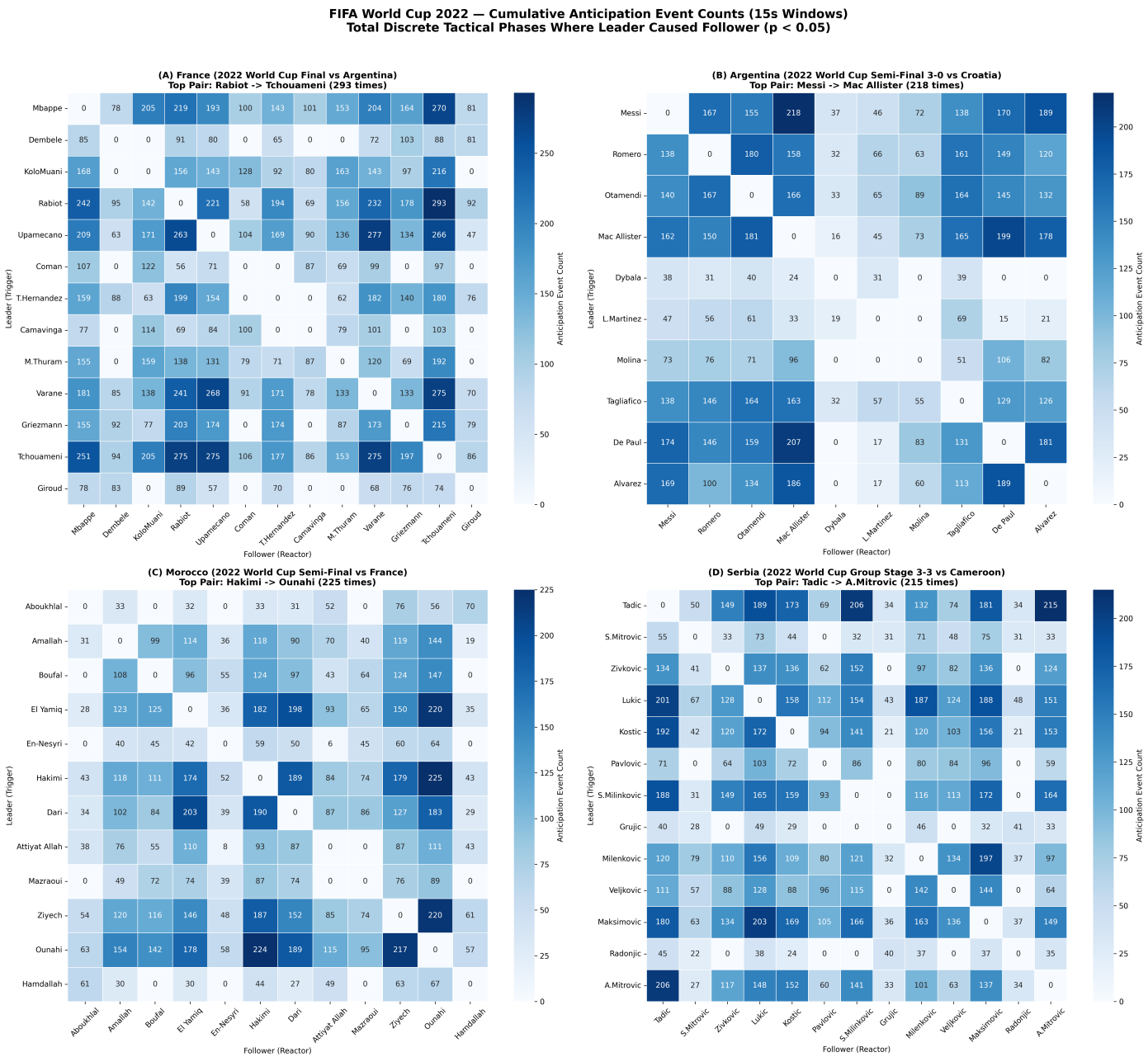


Figure 2: In-Team vs. Cross-Team Anticipation Event Counts (World Cup Final)

Figure 2. Discrete anticipation event counts across 15-second windows in the 2022 World Cup Final between Argentina and France. (A) France In-Team Chemistry: Internal anticipation among French outfield players. (B) Argentina Reads France: How Argentine defenders and midfielders tracked French attackers, led by Cristian Romero anticipating Randal Kolo Muani (256 times) and Ousmane Dembélé (196 times). (C) France Reads Argentina: How French players tracked Argentine attackers, led by Aurélien Tchouaméni reading Alexis Mac Allister (308 times) and Lionel Messi (287 times). Paulo Dybala is included (subbed on in minute 121), while unused substitutes are removed.

